

M cmake/CsmpDependencies.cmake
added a quick line to stop an error message appearing (takes away search for samg library on the default path, and looks only in the correct path.)

M cmake/CsmpPlatform.cmake
Took away -fshort-enums flag which was causing incompatible compilation with libc++ !

M source/includes/integration/finite_elements/**IsoparametricLinearLineElement.h**
Added method interiorNodes , and changed gaussian points

M source/includes/integration/finite_elements/**IsoparametricQuadraticLineElement.h**
Bug Fix with incorrect number of corner and mid point nodes returned! It was just plain wrong.

M source/includes/integration/finite_elements/**IsoparametricQuadraticTriangle.h**
i don't think i changed this file

New!

D source/includes/integration/pde_integrators/IterativeIntegrator_Experimental.h
created class

M source/includes/integration/pde_integrators/PDE_IntegratorExperimental.h
see cpp

D source/includes/integration/pde_integrators/PicardIntegrator_Experimental.h
created class

D source/includes/integration/pde_integrators/StaggeredIntegrator_Experimental.h
do not Pull- its not working / used

M source/includes/integration/pde_operators/numerical/**NumIntegral_BT_D_B_dV.h**
Added method to query LHS from outside class - This is needed for my ExternalLoad Numintegral which is a post processor which takes the x vector and the stiffness matrix and calculates the f vector which resembles the loads on a system. This is very useful to check Neumann Boundaries are applied correctly and is needed to validate three point bending benchmarks and simulate load-displacement curves

D source/includes/integration/pde_operators/numerical/**NumIntegral_NT_n_P_dS.h**
Num int to apply aperture volume change term to displacements. displacements is test operand but pressure is basis operand. This is for LHS and accumulated on a split boundary . UF

D source/includes/integration/pde_operators/numerical/**NumIntegral_NT_n_rhs_P_dS.h**
same as above but RHS version .. . UF

D source/includes/integration/pde_operators/numerical/**NumIntegral_PT_n_N_dS.h**
created num int to apply tractions on split boundary on the LHS. tractions must be scalar and defined in the Intervening element. UF

D source/includes/integration/pde_operators/numerical/**NumIntegral_PT_n_op_n_P_dS.h**
RHS version of above... created num int to apply tractions on split boundary on the RHS. tractions must be scalar and defined in the Intervening element. UF

M source/includes/integration/pde_operators/numerical/
NumIntegral_PT_op_dS_Experimental.h
Applies tractions to split boundary (may be repeat of above but not sure)

M source/includes/integration/solver/**SAMG_Settings.h**
Added ability to set a few more parameters in SAMG

M source/includes/integration/solver/**SAMG_Solver.h**

Added the `iu_vector` needed for calibration along with the ability to input some new settings within SAMG. An important one for my HF simulations is `CSET_UNKNOWN` which can exclude some unknowns from the coarsening process, which was helpful in stabilising the solver at the start of the simulation

M source/includes/integration/solver/**Solver.h**

Method added which can take a `iu_vector` (needed for derived class `SAMG_Solver`)

M source/includes/model/**BoundaryInterface.h**

Added ability to remove faces from a boundary, which is needed to as my split boundary propagates along a boundary, and so the faces which are split to interfaces now must be deleted.

M source/includes/model/**Element.h**

Compile time error fixed, whereby `Assign` method in `FiniteElementPolicy` was being shadowed by `Element::Assign`. So quick fix applied. added bool to ask if quarter point (you won't like this stephan :-D)

M source/includes/model/**Face.h**

Added/modified method giving parent face number based on side of interface

M source/includes/model/**FiniteElementManager.h**

BUG FIX: Added method which re-initialises the object ID such that `CoordinateMatrix()` will always be updated. This was causing a random bug which accessed incorrect elements if the element id's had changed (happens when propagating mesh)!

M source/includes/model/**Index.h**

GCC caught this compile time error - index tracker was not initialised to `nullptr`.

M source/includes/model/**InputDataManager.h**

nothing changed (a comment)

M source/includes/model/**InterFace.h**

Added method to search node on side of interface and replace it with another (used for splitting)

M source/includes/model/**MathOperatorLHS.h**

Math operator expanded to accumulate interface objects and importantly, to receive information regarding which side of interface should the indexing be taken from the node id's! this varies based on the num int you have and so must be decided at Num int level

M source/includes/model/**Matrix.h**

M source/includes/model/**ModelTopology.h**

bug fixes in class regarding surface orientations ... done in cpp file .

M source/includes/model/**RegionInterface.h**

Added method to form region from pointer of element vector

M source/includes/model/**SplitBoundary.h**

Added method to propagate split boundary based on input node to split

Added method which sorts connectivity of middle elements

M source/includes/model/**SplitBoundaryInterface.h**

Added method to propagate split boundary and update model

M source/includes/property_manipulation/**ConstantFactor.h**

Added capacity for vectors

D source/includes/thmc/mechanics/CohesiveTip.h

Application Specific

D source/includes/thmc/mechanics/DisplacementCorrelationTip.h

Application Specific

D source/includes/thmc/mechanics/ExternalLoad.h

Calculates external load on system

M source/includes/thmc/mechanics/Fracture.h

Application Specific

D source/includes/thmc/mechanics/FracturePropagator.h

NOT WORKING

D source/includes/thmc/mechanics/FractureTip.h

Application Specific

D source/includes/thmc/mechanics/HydraulicFractureTip.h

Application Specific

D source/includes/thmc/mechanics/InterFaceApertureVisitor.h

Visitor for updating aperture

D source/includes/thmc/mechanics/J_IntegralTip.h

Application Specific

M source/includes/utilities/**vsetMakers.h**

Added method which creates small mesh with boundary\splitboundary by making vset.

M source/modules/integration/finite_elements/FiniteElement.cpp

nothing changed

M source/modules/integration/finite_elements/FiniteElementPolicy.cpp

M source/modules/integration/finite_elements/IsoparametricLinearLineElement.cpp

M source/modules/integration/finite_elements/IsoparametricQuadraticTriangle.cpp

D source/modules/integration/finite_elements/IsoparametricQuadraticTriangleQuarterPoint.h

M source/modules/integration/finite_elements/VData.cpp

D source/modules/integration/pde_integrators/IterativeIntegrator_Experimental.cpp

M source/modules/integration/pde_integrators/PDE_IntegratorExperimental.cpp

D source/modules/integration/pde_integrators/PicardIntegrator_Experimental.cpp

D source/modules/integration/pde_integrators/StaggeredIntegrator_Experimental.cpp

D source/modules/integration/pde_operators/fe_numerical_integration/ExternalLoad.cpp

M source/modules/integration/pde_operators/fe_numerical_integration/
NumIntegral_BT_D_B_dV.cpp

D source/modules/integration/pde_operators/fe_numerical_integration/
NumIntegral_NT_n_P_dS.cpp

D source/modules/integration/pde_operators/fe_numerical_integration/
NumIntegral_NT_n_rhs_P_dS.cpp

D source/modules/integration/pde_operators/fe_numerical_integration/
NumIntegral_PT_n_N_dS.cpp

D source/modules/integration/pde_operators/fe_numerical_integration/
NumIntegral_PT_n_op_n_P_dS.cpp

M source/modules/integration/pde_operators/fe_numerical_integration/
NumIntegral_PT_op_dS.cpp

M source/modules/integration/pde_operators/fe_numerical_integration/
NumIntegral_SetRHS_to_Zero.cpp
M source/modules/integration/solver/SAMG_Settings.cpp
M source/modules/integration/solver/SAMG_Solver.cpp
M source/modules/interfaces/TextFileIO.cpp
M source/modules/model/BoundaryInterface.cpp
Bug Fix regarding creation of boundary, should be checked ..., but is used .

M source/modules/model/Element.cpp
M source/modules/model/Face.cpp
M source/modules/model/FiniteElementManager.cpp
M source/modules/model/InterFace.cpp
M source/modules/model/MathOperatorLHS.cpp
M source/modules/model/MathOperatorRHS.cpp
M source/modules/model/Matrix.cpp
M source/modules/model/Model.cpp
M source/modules/model/ModelSubDomain.cpp
M source/modules/model/ModelTopology.cpp
M source/modules/model/RegionInterface.cpp
M source/modules/model/SplitBoundary.cpp
M source/modules/model/SplitBoundaryInterface.cpp
M source/modules/property_manipulation/interrelations/ConstantFactor.cpp
M source/modules/property_manipulation/interrelations/CubicLawConductivity.cpp

From here on, we mainly have application specific code

D source/modules/thmc/mechanics/CohesiveTip.cpp
D source/modules/thmc/mechanics/DisplacementCorrelationTip.cpp
M source/modules/thmc/mechanics/Fracture.cpp
D source/modules/thmc/mechanics/FracturePropagator.cpp
not started / finished
D source/modules/thmc/mechanics/FractureTip.cpp
D source/modules/thmc/mechanics/HydraulicFractureTip.cpp
D source/modules/thmc/mechanics/InterFaceApertureVisitor.cpp
D source/modules/thmc/mechanics/J_IntegralTip.cpp
M source/modules/thmc/mechanics/StressesAndStrains2.cpp
M source/modules/utilities/vsetMakers.cpp
D tests/integration/FracturePropagation_VVCase.cpp

Application

D tests/integration/FracturePropagation_VVCase.h

Application

D tests/integration/HydraulicFracturing_VVCase.cpp

Application

D tests/integration/HydraulicFracturing_VVCase.h

Application

D tests/integration/KGD_Crack_Staggered_VVCase.cpp

Application

D tests/integration/KGD_Crack_Staggered_VVCase.h

Application

M tests/integration/KGD_Crack_VVCase.cpp

Application

M tests/integration/KGD_Crack_VVCase.h

Application

D tests/integration/LinearElasticFractureAperture2D_VVCase.cpp

Application

D tests/integration/LinearElasticFractureAperture2D_VVCase.h

Application

M tests/integration/LinearElasticIsotropicDeformation2D_VVCase.cpp

Application

M tests/integration/LinearElasticIsotropicDeformation2D_VVCase.h

Application

D tests/integration/PennyShapedCrack_VVCase.cpp
Application
D tests/integration/PennyShapedCrack_VVCase.h
Application
D tests/integration/SneddonCrackCoupled_VVCase.cpp
Application
D tests/integration/SneddonCrackCoupled_VVCase.h
Application
M tests/integration/SneddonCrack_VVCase.cpp
Application
M tests/integration/SneddonCrack_VVCase.h
Application
D tests/integration/StressIntensityFactorTest_VVCase.cpp
Application
D tests/integration/StressIntensityFactorTest_VVCase.h
Application
M tests/unit/InterFace_Test.cpp
M tests/unit/SplitBoundaryInterface_Test.cpp
M tests/unit/SplitBoundaryInterface_Test.h
M tests/unit/SplitBoundary_Test.cpp
M tests/unit/SplitBoundary_Test.h
M tests/unit/VSet_TestCase.cpp
M tests/unit/VSet_TestCase.h
M tests/unit_tests_development.cpp
M tests/unit_tests_main.cpp